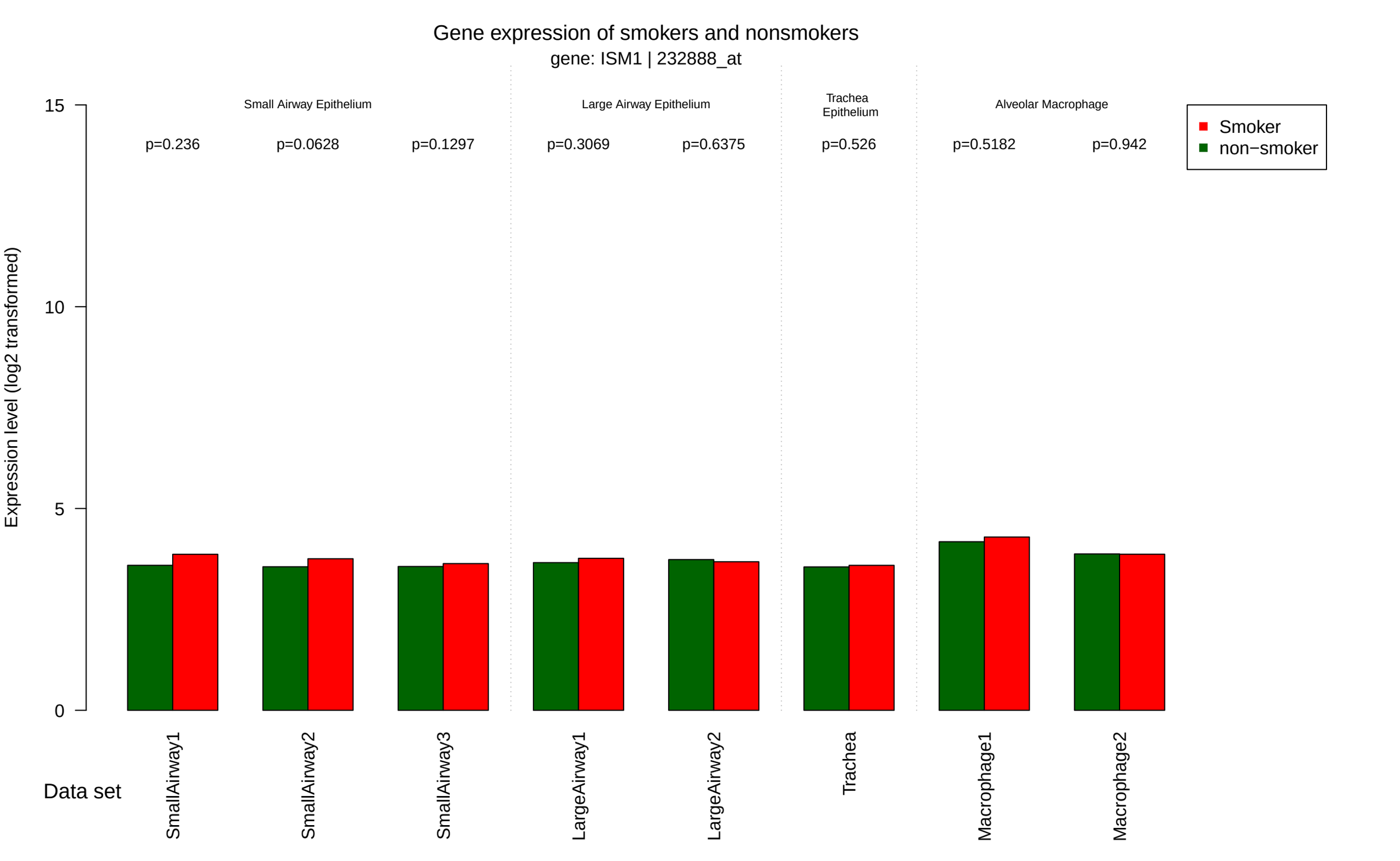
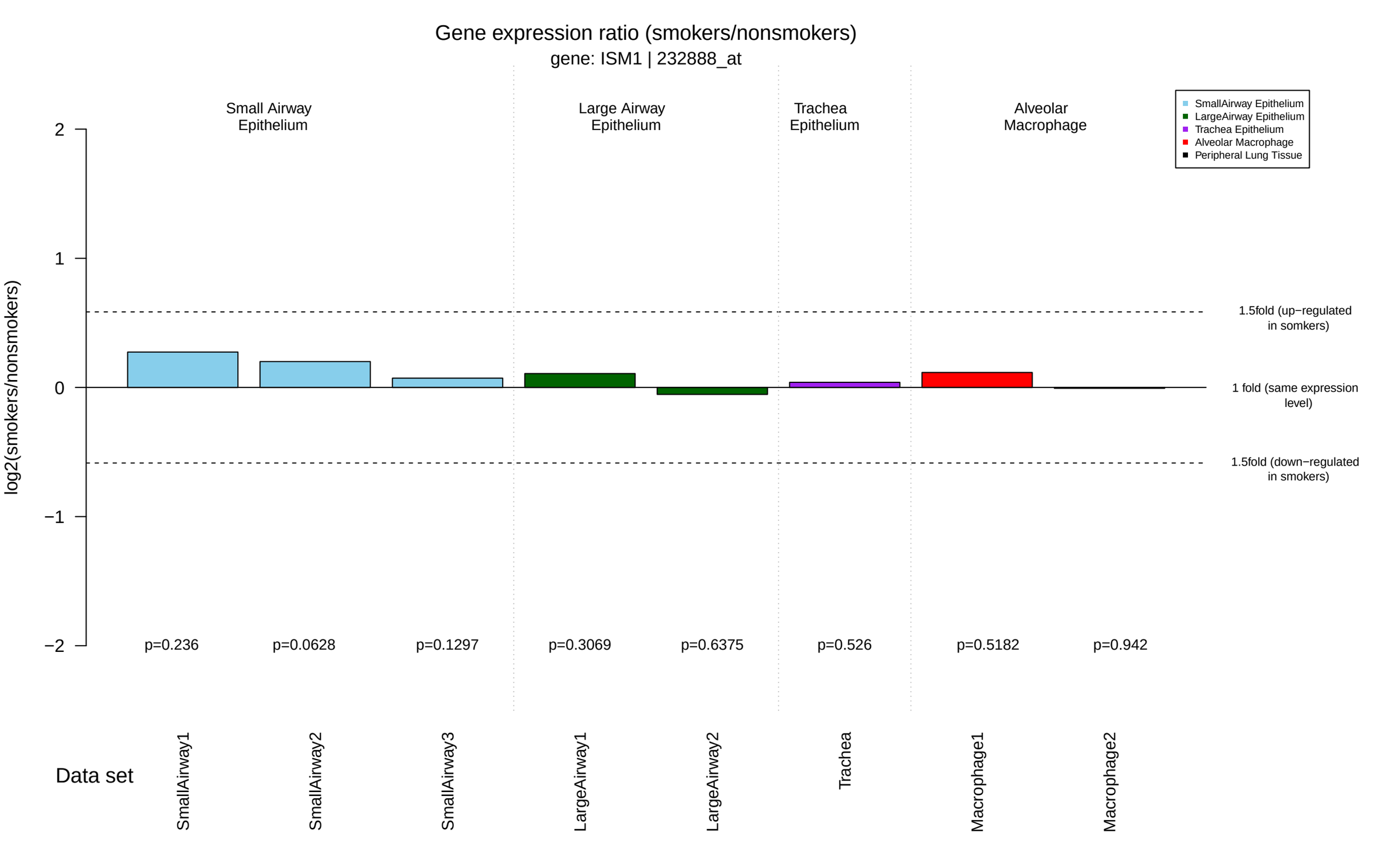
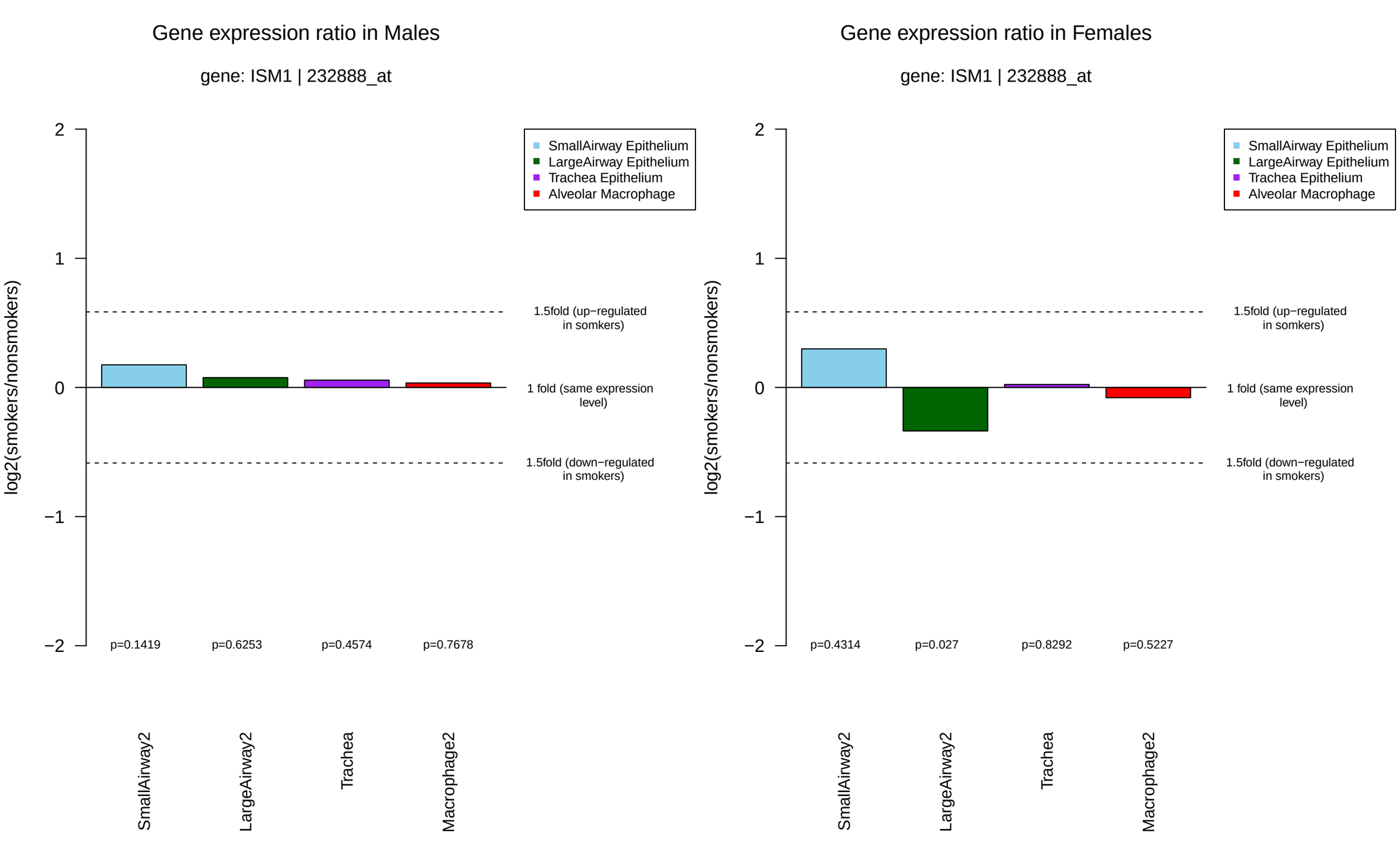
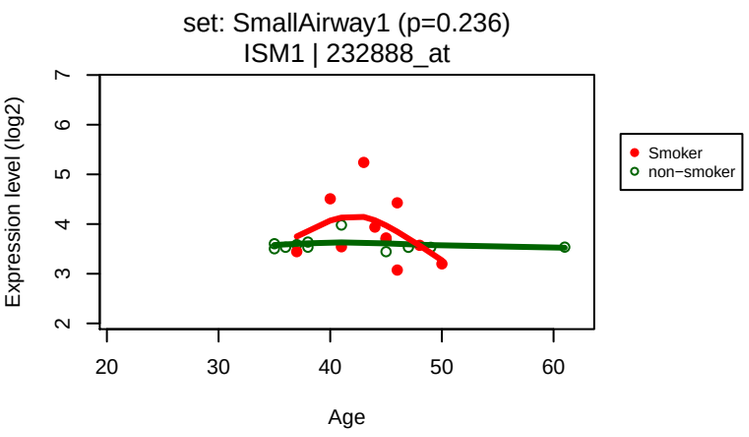
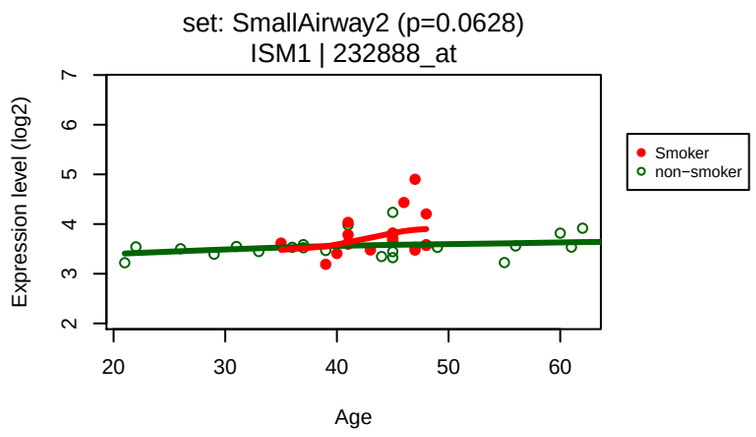
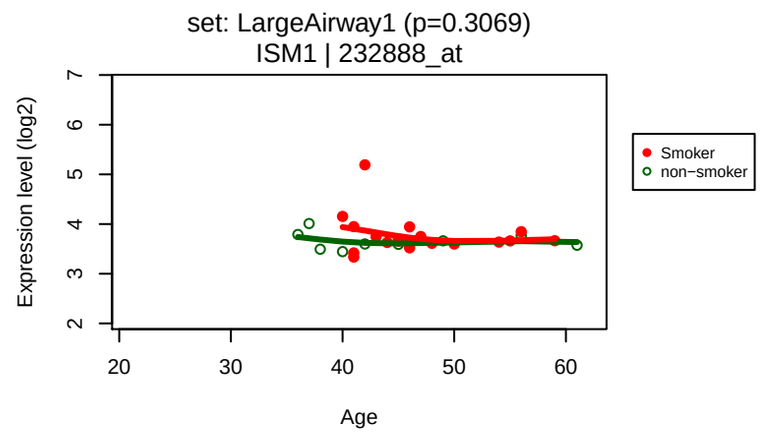

Gene expression of smokers and nonsmokers
gene: ISM1 | 232888_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.236
p=0.0628
p=0.1297
p=0.3069
p=0.6375
p=0.526
p=0.5182
p=0.942
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

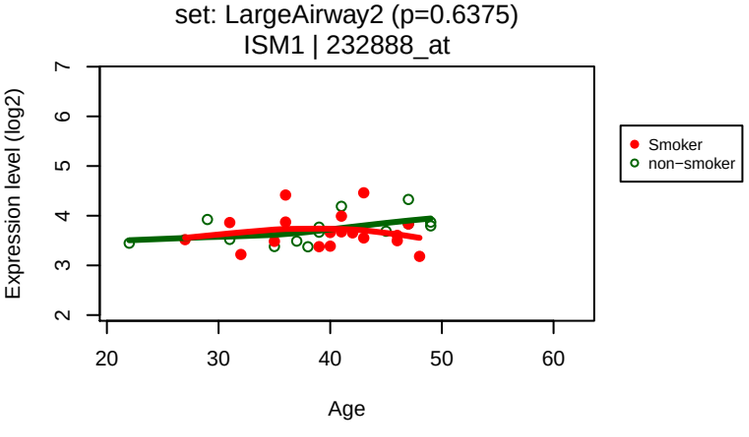

Gene expression ratio (smokers/nonsmokers)
gene: ISM1 | 232888_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.236
p=0.0628
p=0.1297
p=0.3069
p=0.6375
p=0.526
p=0.5182
p=0.942
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

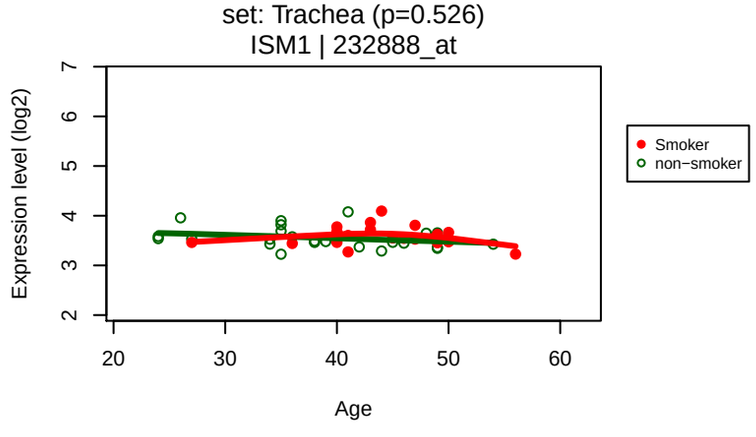

Gene expression ratio in Males
gene: ISM1 | 232888_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1419
p=0.6253
p=0.4574
p=0.7678
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISM1 | 232888_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4314
p=0.027
p=0.8292
p=0.5227
SmallAirway2
LargeAirway2
Trachea
Macrophage2

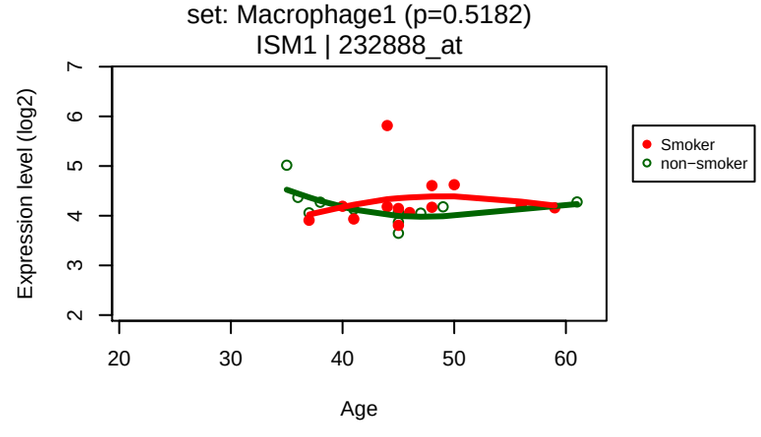

set: SmallAirway1 (p=0.236)
ISM1 | 232888_at
Expression level (log2)
Age
Smoker
non−smoker

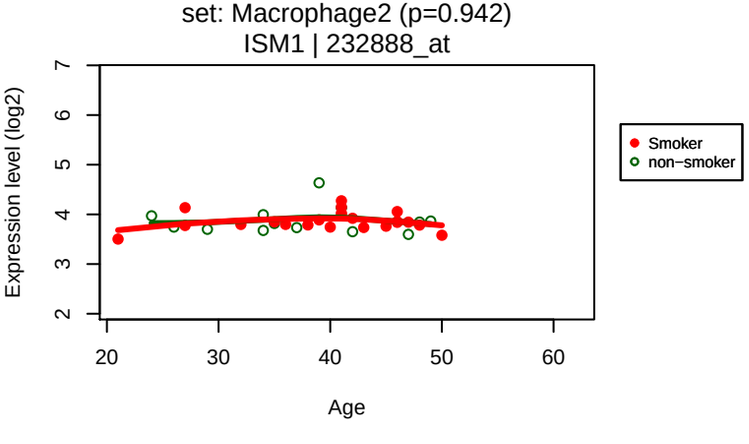

set: SmallAirway2 (p=0.0628)
ISM1 | 232888_at
Expression level (log2)
Age
Smoker
non−smoker

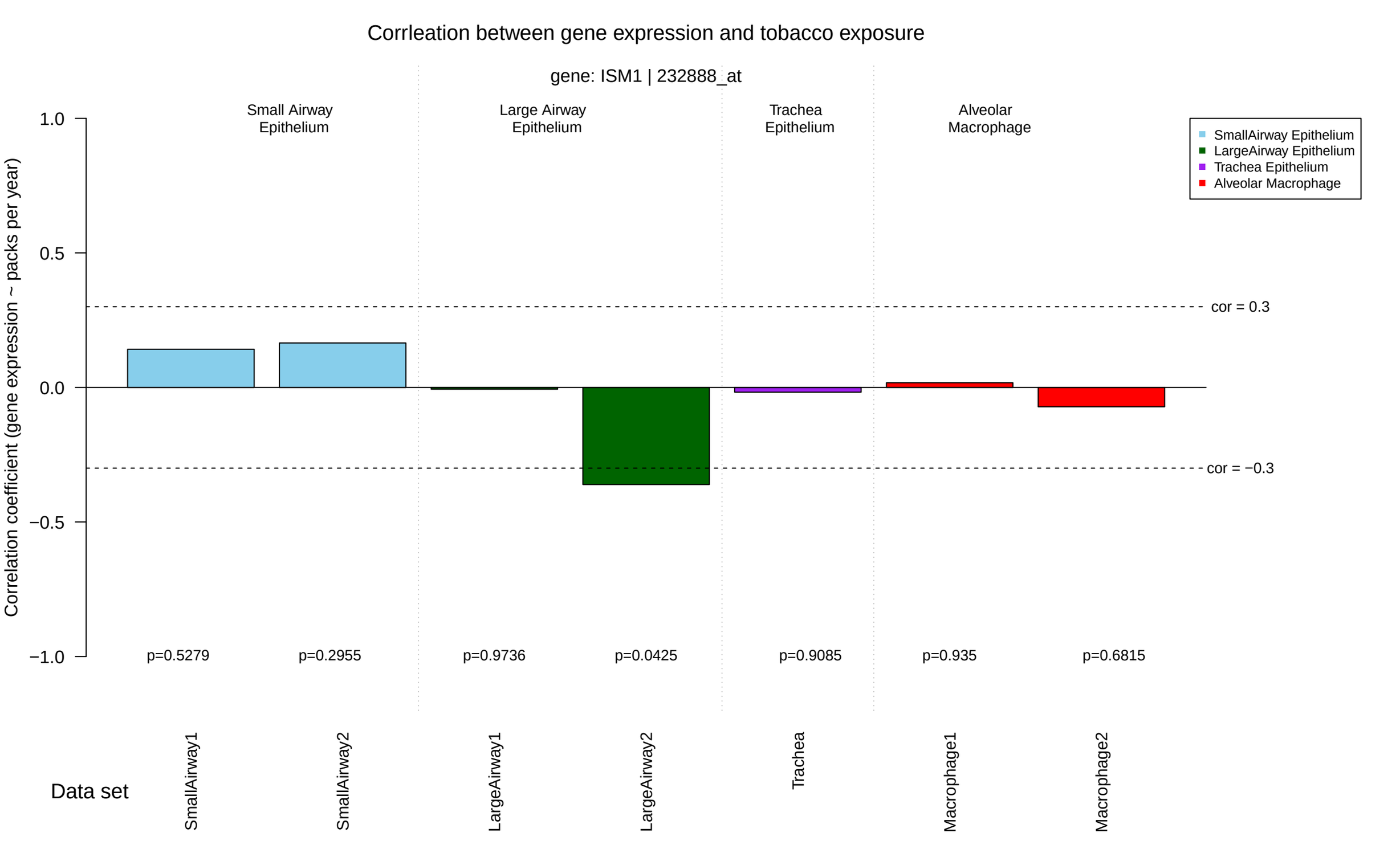

set: LargeAirway1 (p=0.3069)
ISM1 | 232888_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6375)
ISM1 | 232888_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.526)
ISM1 | 232888_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5182)
ISM1 | 232888_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.942)
ISM1 | 232888_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISM1 | 232888_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5279
p=0.2955
p=0.9736
p=0.0425
p=0.9085
p=0.935
p=0.6815
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2